Deep Reinforcement Learning Driven Weakly Supervised Lesion Segmentation in Lung CT Sequence Images

Yexin Lai, Xueyu Liu, Xinyi Liang, Chenyang Rong, Yongfei Wu, *Member, IEEE*, and Wen Zheng

***Abstract*—Accurate and automated segmentation of interstitial lung disease (ILD) lesions from computed tomography (CT) sequence images is essential for quantitatively assessing disease progression and evaluating potential therapeutic strategies. The heterogeneity of ILD lesions and the complexity of manual annotations present significant challenges in achieving precise segmentation. Leveraging the inherent continuity of slice data in CT, where lesion shapes and locations exhibit systematic and progressive changes, we propose a deep reinforcement learning (DRL)-driven weakly supervised ILD lesion segmentation framework, called CT smoother agent (CTSA). Specifically, given an initial mask trained by an arbitrary weakly supervised network, CTSA uses the DRL algorithm to ensure the continuity of the CT sequence by changing the value of pixels in the optical flow (OF) change region, so as to obtain an optimized mask. Subsequently, CTSA uses a dual U-Net to train the constraint segmentation module based on the optimized mask and the initial mask. This process is cycled to achieve the purpose of segmenting ILD lesions. The policy network and segmentation network interact to optimize ILD lesion boundaries and segmentation results simultaneously. We evaluated our model on a dataset comprising 306 ILD patients, and the proposed CTSA improves dice similarity coefficient (DSC) by 9.3%, mean intersection over union (mIoU) by 21.7%, and Hausdorff distance (HD) by 1.522. The experimental results show that our proposed CTSA optimization model can achieve the state-of-the-art (SOTA) performance by learning the contextual information of CT sequences. Our code has been released at: https://anonymous.4open.science/r/CT-Smoother-Agent-1E8A**

***Index Terms*—Computed tomography (CT) sequence image segmentation, deep reinforcement learning (DRL), dual U-Net, interstitial lung disease (ILD), weakly supervised segmentation (WSS).**

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I. Introduction

INTERSTITIAL lung disease (ILD) refers to a variety of diseases that mainly affect the lung interstitium and disrupt the gas exchange process [1], which can lead to potentially life-threatening complications [2]. Early diagnosis is crucial for making treatment decisions. Computed tomography (CT) is generally considered central to the diagnosis of ILDs due to the specific radiation attenuation properties of lung tissue [3]. Therefore, CT image segmentation is of great significance in computer-aided diagnosis (CAD) and can help radiologists quickly locate lesions. However, radiographic assessment of ILD requires considerable experience and expertise, and can vary widely among experienced radiologists [4]. In addition to this, the heavy task of radiological interpretation places a heavy workload on radiologists, and the development of automated quantitative methods may be of important clinical relevance and interest. Furthermore, automated methods can overcome interobserver variability in visual assessment of the extent of ILD [5].

In recent years, deep learning has achieved state-of-the-art (SOTA) performance for many medical image processing tasks [6], [7], [8], [9], [10]. Current effective supervised segmentation methods usually require a large number of high-quality annotations. However, acquiring these detailed annotations is often time-consuming and labor-intensive for radiologists. To mitigate these challenges, researchers have explored various strategies such as data augmentation [11], [12], [13], transfer learning [14], [15], and domain adaptation [16], [17] to reduce reliance on extensive labeled data.

Nowadays, weakly supervised segmentation (WSS) methods, employing minimal annotations such as points and scribbles for generating pseudo-labels, have gained increasing attention [18], [19], [20], [21]. Although these methods reduce the workload of pixel-level annotation, they still require artificial constraints. Among various types of weak annotations for semantic segmentation, image-level class labels have been widely used [22], [23], [24], [25]. However, learning semantic segmentation with image-level label supervision is highly challenging because such labels only indicate the presence of an object class without providing its location or shape. This limitation is particularly critical in medical imaging, where diseases exhibit complex and varied morphology and position, ultimately leading to low segmentation accuracy.

In clinical practice, CT scanning is usually performed through a series of consecutive slices to obtain image infor-

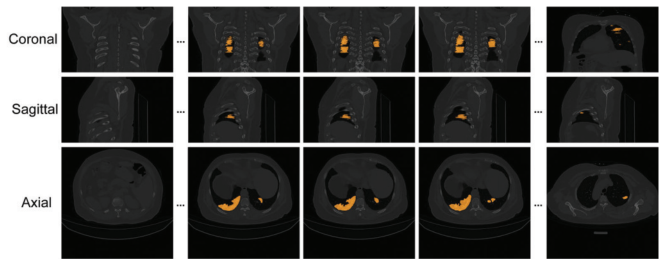

Fig. 1. Three imaging planes of the CT sequence. The figure shows the coronal plane, sagittal plane, and axial plane of the same CT sequence, where the orange part is the lesion.

mation at different levels of the body. In the CT imaging process, the organs in scanned CT images are always highly continuous [26]. From this perspective, we could assume that the lesion region is continuously changed in the CT sequence. Fig. 1 shows three planes of a CT sequence. We can observe that the lesion region is spatially continuous. And the human vision system can identify an object via gradual and intelligent change of the retina fixation [27], which can be formulated as a Markov decision process [28] and solved by deep reinforcement learning (DRL) [29]. Therefore, we advocate the use of DRL to identify the location and boundaries of ILD lesions on each CT slice, with which we can develop an accurate segmentation method for ILD lesions without the use of pixel-level labels.

In this article, we propose a DRL-based cycle ILD lesion segmentation model, called CT smoother agent (CTSA), which performs WSS on CT sequence images using only image-level labels. First, we use image-level label WSS algorithms, such as vision transformer (ViT)-MSIL [30], to obtain the initial masks. Then, we use CT slices and initial masks to train the DRL module and constraint segmentation module (e.g., U-Net). In the DRL module, we introduce optical flow (OF) detection to identify the regions with large differences in morphology and position between adjacent slices, and take actions (i.e., increase or decrease of pixel value) to obtain optimized masks. Then, the optimized masks and the initial masks are jointly input into the constraint segmentation module in a certain proportion for training to obtain enhanced masks. Finally, enhanced masks are input into the DRL module again for optimizing, and obtain the final optimization results through the cycle. The main contributions are summarized as follows.

1) We propose CTSA, a DRL-based cyclic ILD lesion segmentation model that optimizes segmentation by learning contextual information from CT sequences.
2) By designing specific actions and reward functions, DRL captures CT sequence context, improving ILD lesion segmentation and overall performance.
3) We propose a dual U-Net, which constrains training using initial and optimized masks to enhance segmentation accuracy.

II. Related Works

A. WSS Methods for CT Images

Semantic segmentation with weak supervision can reduce the dependence on fully supervised labels, and more and more work has applied weakly supervised learning to CT image segmentation. Laradji et al. [31] propose a consistency-based (CB) loss function to segment COVID-19 infections using point annotations. Liu et al. [32] propose a novel weakly supervised method for the segmentation of COVID-19 infections in CT slices with scribble supervision. Sun et al. [33] propose a local self-coherence mechanism to accomplish label propagation based on lesion area labeling characteristics for weak labels that cannot offer comprehensive lesion areas, hence forecasting a more complete lesion area. Lu et al. [34] design a lightweight weakly supervised convex-hull-based D-block network (CHDNet) to segment the lesions of COVID-19 using point annotations. Guo et al. [35] propose a weakly supervised abnormality segmentation algorithm based on the self-supervised equivariant attention mechanism (SEAM) model and compose of self-supervision and autovariant attention mechanisms. This method can distinguish lesions of ILD and segment abnormal areas. However, due to the lack of consideration of the contextual relationships of CT sequences, there are false detection and missed detection of lesions in different slices of the same sequence, resulting in poor segmentation results.

B. DRL-Based Medical Image Segmentation

Reinforcement learning (RL) is a learning method that aims to make the agent learn the optimal policy through the process of trial and error through interaction with the environment. Mnih et al. [36] designed a deep neural network to perform the action-value function approximation in Q-learning and proposed the DRL algorithm for the first time. Since then, many methods based on DRL have been applied in medical image processing. Yang et al. [37] propose a DRL search method to optimize the training policy for 3-D medical image segmentation, which improves the performance of the baseline model. Man et al. [38] first apply the deep Q-learning (DQN) to the identification of the bounding box of each pancreas, and then use a modified U-Net to segment the pancreas in cropped CT images. Li and Xia [39] propose a deep reinforcement learning-based lymph node segmentation (DRL-LNS) model to segment lymph nodes on CT images using response evaluation criteria in solid tumor (RECIST) annotation. However, these works do not utilize the contextual knowledge between CT slice sequences. Ding et al. [40] propose an RL-based brain tumor segmentation network (RLSegNet) to transform the brain tumor segmentation process into a series of decision problems by exploiting the contextual relationships between magnetic resonance imaging (MRI) images. However, the optimization process of this method depends on the quality of the masks. If the mask is wrong, it will affect the training.

III. Methodology

In this section, we describe our proposed CTSA for ILD lesion identification and segmentation, as shown in Fig. 2. The proposed model consists of an initial mask generation (IMG) module (i.e., ViT-MSIL), a DRL module, and a constraint segmentation module. The model is trained in a weakly supervised manner, which is mainly divided into three steps: 1) the

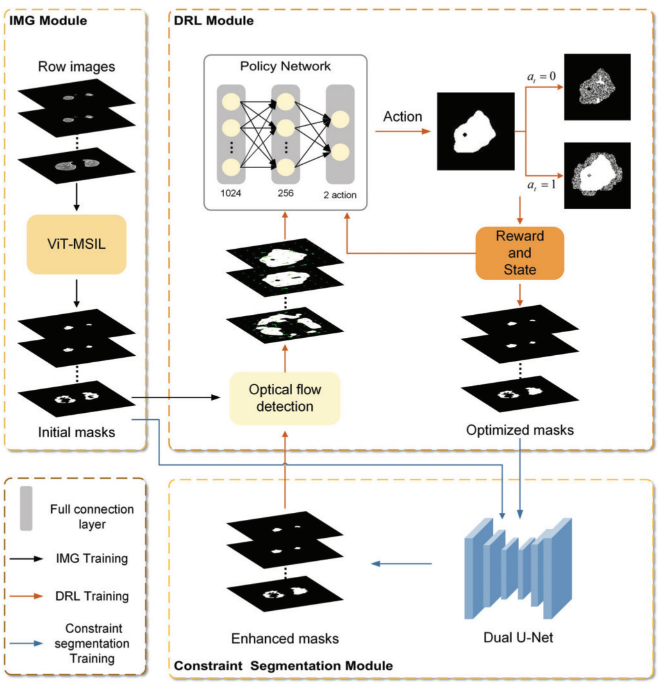

Fig. 2. Overall pipeline of the proposed CTSA framework. The model is mainly divided into three parts: the first part is the IMG module, which is used to segment the ILD lesions from CT, and the second part is the DRL module, where we combine OF detection and DRL to learn contextual information between CT slices. The third part is the constraint segmentation module, which can better optimize segmentation results by training optimized masks and initial masks.

ILD lesions are segmented using ViT-MSIL to obtain the initial masks; 2) the initial mask is used to train the DRL module with OF detection, and the policy network is expected to find the optimal segmentation results of ILD lesions on consecutive slices and obtain a optimized mask; and 3) the dual U-Net is trained with initial mask and optimized mask to obtain enhanced mask. The cooperation between the policy network and dual U-Net leads to a gradually improving the ILD lesion segmentation mask until the stop action is performed or the policy network reaches the maximum search steps. In Sections III-A–III-D, we would describe all these components in detail.

A. IMG Module

We use the ViT-MSIL [30] as the IMG module to obtain the initial mask based on image-level labels. This algorithm constructs an end-to-end multiple-instance learning (MIL) method to realize the automatic learning of superpixel-level segmentation from image-level labels. Specifically, we use the Felzenszwalb (Felz) [41] algorithm to generate superpixel instance, and then put these instances into the classifier trained with ViT [42] as the backbone for classification. Finally, the initial mask is obtained.

B. DRL Module

A DRL module consists of an agent that infers an action and the environment that gives a reward reflecting the significance of the action and a new state to the agent. We aim to train

Algorithm 1 DRL Training Process

Input: CT image sequence $I_i^{(1,2,\ldots,k,\ldots,n)}$, denotes i-th CT sequence contains n CT slices
Parameter: Max step; Max epoch;
Output: Q-values

Initializing the Q-values, epoch, step.
while Epoch < Max epoch **do**
  Calculation of optical flow anomaly regions $\mathcal{A}$.
  Initializing the step.
  while Step < Max step **do**
    Randomly select a pixel (x, y) from $\mathcal{A}$
    Empty reward.
    Action list = δ_{a_t}.
    Select an action from the action list.
    Add the reward r_t based on the reward function.
    Update Q-values
    Step + = 1.
  end while
  Epoch + = 1.
end while
return solution

our goal is to train the agent to learn policies that change the abnormal regions of lesions to the best shape and location, which boosts the dual U-Net segmentation. The algorithm is described in Algorithm 1. We now introduce the main components of the DRL module.

1) State: In the DRL framework, the state s_t represents the agent's observation of the environment at time step t. In this task, the state corresponds to the pixel matrix of the current image. Assuming the input image at time t is I_t, the state s_t can be defined as

$$s_t = I_t \tag{1}$$

where I_t^n is a 2-D matrix representing the image at time t. Each pixel value is a real number, $I_t^n \in R^{H\times W}$, where H and W are the height and width of the image, respectively. This definition allows the agent to have a complete representation of the current CT slice, enabling it to make informed decisions based on the spatial distribution of pixel intensities.

2) Action: Before defining the action, we introduce the OF detection module. The OF detection is used to calculate the motion vector field between two consecutive slices, which is denoted as $F(I_i^k, I_i^{k+1})$

$$F\left(I_i^k, I_i^{k+1}\right) = (u, v) \tag{2}$$

where I_i^k refers to the kth slice in the ith CT sequence. $u(x, y)$ and $v(x, y)$ denote the displacement of the pixel in the x- and y-directions, respectively.

The OF detection extracts motion information and identifies possible abnormal changes in the image. The magnitude of OF is denoted as $M(x, y)$

$$M(x, y) = \sqrt{u^2 + v^2} \tag{3}$$

which reflects the amount of motion of pixel (x, y) in the image.

Abnormal regions are identified by the standard deviation of the OF

$$\mathcal{A} = \begin{cases} 1, & M(x, y) > \mu + \sigma \cdot \tau \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $\mathcal{A}(x, y)$ denotes whether the pixel (x, y) belongs to an abnormal region, μ and σ are the mean and standard deviation of the OF magnitude, respectively, and τ is the hyperparameter controlling the threshold. In our experiments, τ is empirically set to 2, which effectively identifies regions of significant deviation. This ensures that only the regions with significant differences are processed, allowing the agent to focus on the most relevant areas.

The action a_t is defined as the operation performed by the agent in state s_t. After detecting abnormal regions, the agent will adjust the pixel values of these regions based on the selected action a_t

$$I_{t+1}(x, y) = \text{clamp}\left(I_t(x, y) + \delta_{a_t} \cdot \mathcal{A}(x, y) \cdot R(x, y), 0, 1\right) \tag{5}$$

$$\delta_{a_t} = \begin{cases} +1, & a_t = 1 \\ -1, & a_t = 0 \end{cases} \tag{6}$$

where clamp$(\cdot, 0, 1)$ restricts the result to the range $[0, 1]$. This ensures that the pixel values remain within a valid range for image representation. $I_t(x, y)$ represents the pixel value at position (x, y) at time step t, and $I_{t+1}(x, y)$ is the pixel value adjusted by the action a_t. The random variable $R(x, y)$ is sampled from $[0, 0.3]$ to introduce randomness, which prevents the agent from converging to a premature local optimum. This randomness helps the agent explore different possible adjustments rather than always following a deterministic path, thus increasing the robustness of the learned policy.

3) Reward: The reward function r_t provides feedback to the agent after performing the action a_t at time t. In this work, we assume that the shape, size, and position of lesions in consecutive slices should exhibit consistency. Therefore, the reward function is designed based on the dice similarity coefficient (DSC) between two adjacent slices. The reward function is defined as follows:

$$r_t = \begin{cases} +1, & 0 < \left|D\left(I_i^k, I_i^{k+1}\right) - D\left(I_i^{k-1}, I_i^k\right)\right| \leq D_{\text{init}} \\ -1, & \text{otherwise} \end{cases} \tag{7}$$

where D_{init} is the initial average DSC difference between two adjacent slices, which is used to measure the similarity change before and after adjustments. The reward value of $+1$ encourages adjustments that lead to consistent segmentation results between consecutive slices, while -1 penalizes significant deviations. This setup helps the agent learn a policy that maintains the continuity of the segmentation across the CT sequence, which is crucial for clinical diagnosis.

4) Policy Network: The policy network takes the input state s_t and outputs the value of each action (Q-value). In this work, the policy network is a deep neural network whose structure can be expressed as follows:

$$Q(s_t, a_t; \theta) = f(s_t; \theta) \tag{8}$$

where θ represents the parameters of the network and $f(\cdot)$ denotes the function of the neural network. The structure of this network can be designed with multiple convolutional layers to effectively capture spatial features of the input state, enabling it to make more informed decisions. The agent selects the action that maximizes Q-value

$$a_t = \arg\max_a Q(s_t, a; \theta). \tag{9}$$

With gradient descent, the network parameter θ is updated by minimizing the following loss function:

$$L(\theta) = E\left[\left(r_t + \gamma \max_{a'} Q\left(s_{t+1}, a'; \theta\right) - Q(s_t, a_t; \theta)\right)^2\right] \tag{10}$$

where γ is the discount factor that balances the tradeoff between current and future rewards. Typically, γ is set in the range $[0, 1]$. A higher γ places more emphasis on future rewards, encouraging the agent to consider long-term benefits, while a lower γ focuses on immediate rewards.

C. Constraint Segmentation Module

We propose a dual U-Net as our segmentation network in the constraint segmentation module, inspired by the Siamese network [43] and I2Unet [44]. The dual U-Net consists of two identical U-Net structures that share encoder weights and utilize a common decoder. Specifically, when the image is fed into the training network, one of the U-Net structures is trained with the initial mask, while the other U-Net is trained with the optimized mask optimized by the DRL module. The purpose of training with the initial mask constraint is to effectively prevent the issue of full black pixels that may occur during the cyclic training process, thereby ensuring the stability of model training.

D. Inference Stage

In the inference stage, we first used the WSS algorithm ViT-MSIL to obtain initial masks. Then, based on the policy learned by the DRL model, the optimization decision is executed step by step. At each decision step, the agent selects the best action according to the current image state and updates the optimization region until the termination condition is satisfied. Different from the training stage, the model does not update the policy in the inference process, but directly executes the optimal policy to complete the optimization task efficiently and stably.

IV. Experiments and Results

A. Dataset Description

1) Internal Dataset: The ILD patient data used in this article are all obtained from the Radiology Department Database of Shanxi Bethune Hospital, which do not require informed consent from patients due to its retrospective nature. This study is conducted in accordance with the principles of the Declaration of Helsinki.

The dataset for this article was collected from March 2018 to August 2021, which consists of 306 ILD (positive) and 200 non-ILD (negative) patients. The ratio of male–female patients with ILD was 4:5, and the age ranged from 22 to 82 years, with a mean age of 61 years. The distribution of the dataset

TABLE I
DETAILS OF DATASET DISTRIBUTION

Division	Type	#Patients	#Images
Train	Positive	150	20716
	Negative	130	20260
Test	Positive	156	33213
	Negative	70	10200

TABLE II
RESULTS OF ABLATION EXPERIMENT

Module			DSC (%)	mIoU (%)	HD	PA (%)
IMG	DRL	Constraint				
✓			85.6	60.5	8.2	95.0
✓	✓		86.3	59.0	8.6	95.8
✓	✓	✓	**94.9**	**82.2**	**6.7**	**99.0**

is shown in Table I, the training and test sets do not cross and conform to independent and identically distributed, and the resolution of each image is 512×512.

Based on our previous work [45], the ground truth (GT) of the dataset are pixel-level annotations, which were provided by two radiologists: 1) if the overlap between the two lesion regions labeled by the two radiologists is larger than 80%, we use the intersection of the two regions as the GT and 2) if the overlap is less than 80%, or if a lesion region is annotated by one radiologist but ignored by another, a third senior radiologist will step in to help determine whether the region is diseased or not. In addition, artifacts are usually included in medical images, and to avoid affecting the iterative learning process, we use GCPANet [46] to preprocess the data and keep only the lung parenchyma region.

2) External Dataset: We use two public datasets, the COVID-19 Collection dataset [47] and LIDC-IDRI [48], as external dataset to verify the effectiveness of our proposed method. The COVID-19 Collection dataset contains 3-D CT volumes of 20 COVID-19 patients and pixel-level labels of lesions. LIDC-IDRI is a widely used lung nodule segmentation dataset with 1018 patient CT slices. For each patient image, a two-stage diagnostic annotation was performed by four experienced chest radiologists. In the first stage, each physician independently diagnosed and labeled the lesion location. In the second stage, each physician independently reviewed the annotations of the other three physicians and gave their own final diagnosis.

B. Evaluation Metrics

We employ DSC, mean intersection over union (mIoU) [49], Hausdorff distance (HD) [50], and pixel accuracy (PA) to evaluate segmentation performance. DSC measures the similarity between the prediction and GT. When DSC value closes to 1, it indicates that the prediction is highly overlapping with GT, while a value of 0 means no overlap. mIoU evaluates the overlap between the predicted and GT regions by computing the intersection area divided by the union area. DSC and mIoU are the standard metric of semantic segmentation. HD is defined as the maximum distance between the edge of prediction and GT, which can evaluate the edge prediction accuracy. A smaller HD value indicates that the segmentation boundary is closer to the true boundary. PA is a metric used in semantic segmentation to measure the ratio of correctly predicted pixels to the total number of pixels in the dataset. It evaluates how well a model classifies each pixel in an image. These formulations related to the evaluation metrics are defined as follows:

$$\mathrm{DSC} = \frac{2\,|S_A \cap S_B|}{|S_A| + |S_B|} \tag{11}$$

$$\mathrm{mIoU} = \frac{1}{n}\sum \frac{|S_A \cap S_B|}{|S_A| + |S_B| - |S_A \cap S_B|} \tag{12}$$

where S_A denotes the GT, S_B denotes the segmentation results via our method, and n denotes the number of classes

$$\mathrm{HD} = \max\left(h\left(S_A, S_B\right), h\left(S_B, S_A\right)\right) \tag{13}$$

with

$$h = \max_{a \in S_A}\left\{\min_{b \in S_B} \|a - b\|\right\} \tag{14}$$

$$h = \max_{b \in S_B}\left\{\min_{a \in S_A} \|b - a\|\right\} \tag{15}$$

where $\|\cdot\|$ is the normal form of distance between point sets S_A and S_B

$$\mathrm{PA} = \frac{\sum_{i=0}^{n} p_{ii}}{\sum_{i=0}^{n}\sum_{j=0}^{n} p_{ij}} \tag{16}$$

where n is the total number of classes, or $n+1$ including the background. p_{ii} represents the total number of pixels of true pixel class i that are predicted to be of class i, that is, how many pixels are the total number of pairs for pixels of true class i. p_{ij} represents the total number of pixels of true pixel class i that are predicted as class j; in other words, how many pixels of class i are misclassified as class.

C. Implementation Details

All experiments are implemented on the PyTorch framework and performed on an NVIDIA Tesla V100 32 GB GPU and an Intel[1] Core[2] i7-8700 at 3.2 GHz. The CTSA overall training is 1000 epochs, in which the DRL module is trained for 100 epochs, the Adam optimizer is used to train weights with a decay of $1e^{-3}$ and a fixed learning rate of $1e^{-3}$, and the discount factor $\gamma = 0.9$. And the dual U-Net is randomly initialized and optimized by the stochastic gradient descent (SGD) algorithm with a learning rate of $1e^{-5}$, a momentum value of 0.9, and a maximum epoch of 100.

D. Ablation Experiment

To verify the effectiveness of the proposed method and configuration, we conducted ablation experiments, and the results are shown in Table II and Fig. 3. The experiments break

[1]Registered trademark.
[2]Trademarked.

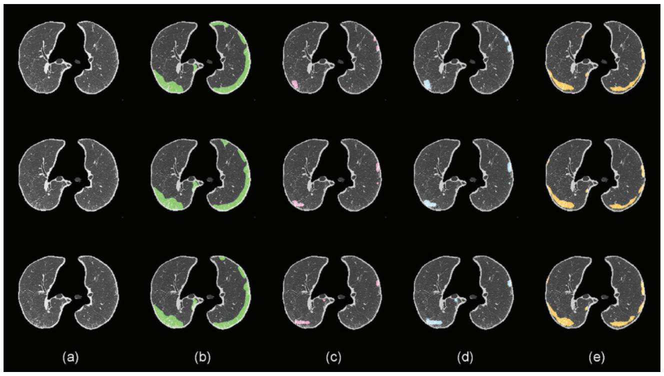

Fig. 3. Visualization results of ablation experiments. Here are three consecutive CT slices of ILD patients. (a) Preprocessed image. (b) GT. (c) With WSS. (d) With WSS and DRL. (e) CTSA.

TABLE III
WHETHER THE DRL MODULE CONTAINS OF DETECTION

	DSC (%)	mIoU (%)	HD	PA (%)
w/o OF	67.7	52.4	9.0	90.1
w/ OF	**94.9**	**82.2**	**6.7**	**99.0**

down each component of the model to assess the contribution of each configuration to the final segmentation performance.

The experimental results show that optimizing the WSS using only the DRL module can also improve the segmentation performance, but not significantly. This is mainly because the WSS method we used only relies on image-level labels, and the initial masks generated are often inaccurate. These inaccurate initial masks directly affect the DRL module's training process, making it difficult for the DRL module to learn effective features from these imperfect pseudo-labels, thus limiting the optimization effect of the module. However, when we introduced the dual U-Net structure, the segmentation results were significantly improved. The dual U-Net module effectively constrains the CTSA's training by combining the optimized masks from the DRL module with the initial masks generated by the WSS method. Specifically, the dual U-Net leverages the information from both mask sources and trains the model collaboratively, helping it learn more accurate features of the lesion areas, which, in turn, improves segmentation accuracy. This approach not only ensures the stability of the training process, but also enhances the CTSA's performance in lesion segmentation tasks by improving information fusion.

1) OF Anomaly Detection: We conducted experiments to evaluate the necessity of including OF anomaly detection in the DRL module. The results shown in Table III indicate that without OF anomaly detection, the DRL module performs negative optimization. We believe this is due to the lack of constraints on abnormal regions, which leads to random actions by the DRL module. Additionally, in CT images, the background occupies a large proportion, which increases the difficulty of optimization. OF anomaly detection can identify abnormal regions in the region of interests (ROIs) of adjacent slices, and taking actions in these areas significantly improves both optimization efficiency and accuracy.

TABLE IV
PERFORMANCE OF TWO FSS ALGORITHMS

Constraint Segmentation Module	DSC (%)	mIoU (%)	HD	PA (%)
Single U-Net	58.1	41.6	10.5	91.7
Dual U-Net	**94.9**	**82.2**	**6.7**	**99.0**

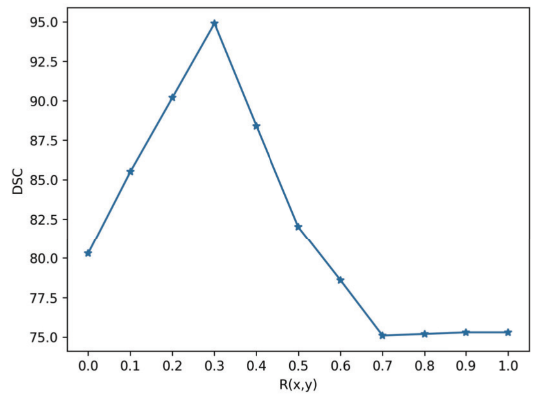

Fig. 4. Line plot of DSC change as $R(x, y)$ changes.

2) Constraint Segmentation Module: To verify the rationality of using the dual U-Net as a constraint segmentation module, we evaluated the two fully supervised segmentation (FSS) algorithms, the single U-Net and the dual U-Net. As shown in Table IV, training a single U-Net with only optimized masks did not work well. This is because the role of the DRL module is to make adjacent slices as similar as possible, and in extreme cases, optimized masks with all black backgrounds may be obtained. Therefore, we need to introduce initial masks to constrain the constraint segmentation module.

3) Random Variable $R(x, y)$: We conducted ablation experiments for random variables $R(x, y)$ of Section III-B2 in the range of [0, 1] with a step size of 0.1. The results are shown in Fig. 4. It is obvious from the figure that with the change of $R(x, y)$, the DSC increases first and then decreases, reaching a peak at 0.3, so we choose the range of $R(x, y)$ between [0, 0.3].

E. Comparative Experiment

1) Results on Internal Dataset: In order to demonstrate the efficacy of our method, we conduct a series of comparative experiments. As can be seen from Table V, our proposed CTSA optimization model can significantly improve the segmentation accuracy compared to other methods. Comparing WSS methods relied on image-level annotation (online attention accumulation (OAA) [51] and Group-WSSS [52]), our method significantly outperforms those methods. These methods perform well on natural images because the segmented objects in natural images often have regular shapes and boundaries, but they are not suitable for medical images with blurred lesion edges. Although the MIL method based on WSS (DWS-MIL [53] and SA-MIL [54]) can perform pixel-level segmentation, both of these methods segment cancer areas on pathological images. Due to the obvious difference

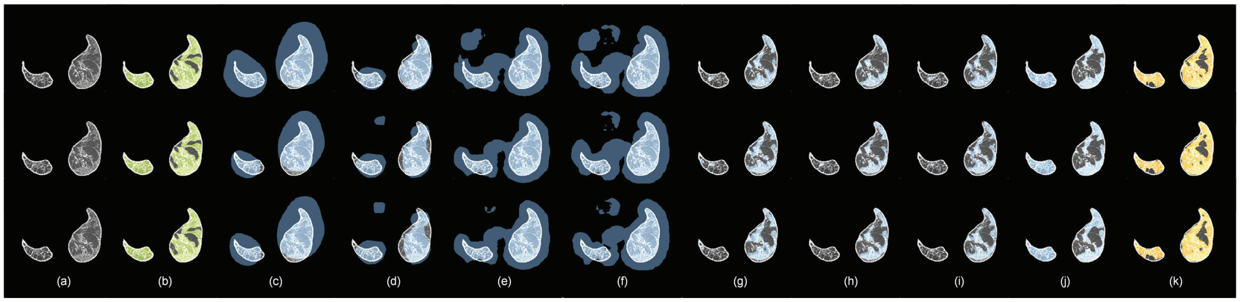

Fig. 5. Visualization results of comparative experiment. (a) Preprocessed image. (b) GT. (c) OAA. (d) Group-WSS. (e) DWS-MIL. (f) SA-MIL. (g) U-Net. (h) Deeplabv3+. (i) U-Net++. (j) P-Net. (k) CTSA.

TABLE V
COMPARISONS WITH WSS METHODS AND FSS METHODS

Supervised	Methods	DSC (%)	mIoU (%)	HD	PA (%)
WSS	OAA [51]	37.2	5.9	22.6	80.1
	Group-WSSS [52]	55.4	26.0	13.5	82.4
	DWS-MIL [53]	62.7	21.7	14.3	80.0
	SA-MIL [54]	61.5	20.5	14.7	81.6
FSS	U-Net [6]	69.6	35.0	9.6	96.6
	Deeplabv3+ [7]	72.4	38.9	9.5	96.3
	U-Net++ [55]	75.4	60.9	8.5	96.1
	P-Net [56]	90.6	80.3	**6.1**	98.4
Ours	CTSA	**94.9**	**82.2**	6.7	**99.0**

TABLE VI
SEGMENTATION PERFORMANCE ON EXTERNAL DATASET

Dataset	Methods	DSC (%)	mIoU (%)	HD	PA (%)
COVID-19	U-Net [6]	76.2	62.6	3.4	99.4
	U-Net++ [55]	80.1	68.2	4.7	98.9
	Deeplabv3+ [7]	79.7	67.1	3.3	97.3
	Inf-Net [57]	75.7	67.3	**2.9**	97.7
	COPLE-Net [58]	69.0	56.1	4.6	98.9
	CTSA	**85.0**	**76.5**	3.0	**99.5**
LIDC-IDRI	U-Net [6]	81.5	69.7	3.4	99.5
	U-Net++ [55]	78.4	65.3	3.7	99.5
	Deeplabv3+ [7]	68.0	52.1	3.6	99.4
	CTSA	**88.0**	**79.0**	**2.9**	**99.7**

between pathological images and the CT images we used, these methods are not effective on our dataset. Compared with the FSS methods, our proposed optimization method surpasses full supervision to reach the SOTA.

The segmentation visualization results are shown in Fig. 5. Fig. 5(a) shows three consecutive CT slices, and it can be seen from the GT that the appearance of the lesion in the CT sequence is continuous. As DWS-MIL and SA-MIL need to cut the photograph into patches for training, the size of the patch is difficult to accurately match the size of the lesion, so this situation will occur in Fig. 5(c)–(f). Fig. 5(i) and (j) shows the results for FSS methods, which is also relatively continuous, but with false negative. However, our proposed CTSA can not only preserve the continuity of the lesion but also segment it accurately.

2) Results on External Dataset: Comparative experiments show that CTSA has excellent performance on different datasets. Table VI presents the quantitative results of comparative experiments on external datasets. It can be seen that all the evaluation indicators of our proposed CTSA are better than the baseline. These excellent results highlight the effectiveness of CTSA in optimizing lesion segmentation in CT sequences, taking full advantage of the continuity of CT sequences and the correlation between adjacent slices.

The visualization results of external datasets are shown in Figs. 6 and 7, where the photographs are three consecutive lung CT images, respectively, and it can be clearly seen that our proposed CTSA is significantly better than other methods and the results are continuous.

TABLE VII
RESULTS OF RUNTIME ANALYSIS

GPU Model	Initial masks generation time (s)	Optimization time (s)	Total time (s)
NVIDIA Tesla V100	18.5	45.2	63.7
NVIDIA RTX 3090	22.3	52.7	75.0
NVIDIA RTX A4500	24.8	55.4	80.2

F. Runtime Analysis

To verify that our method is still very efficient and practical in resource-constrained environments, we conduct runtime analysis experiments on three different GPUs. The experimental results are shown in Table VII, where the total running time remains at a reasonable 80.2 s even on a workstation-level NVIDIA RTX A4500. These results show that our method is not only scalable but also adaptable to various hardware configurations, including environments with limited computational resources.

V. DISCUSSION

The segmentation of medical sequences plays an important role in ILD diagnosis, but the current mainstream methods only process each CT image, ignoring the fact that the scanned objects are always highly continuous. Moreover, the segmentation methods based on deep learning often rely on a large number of labeled samples. However, due to the huge demands on clinical expertise and time, it is often impractical to collect a large number of annotations, especially for high-dimensional medical sequences. In this work, we exploit the continuity of

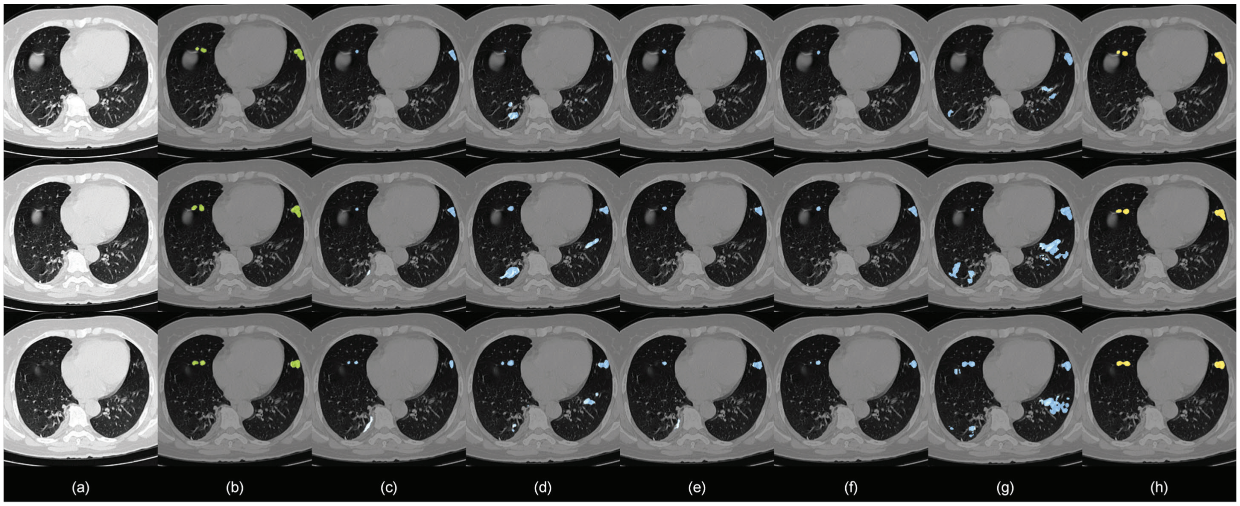

Fig. 6. Visualization results of COVID-19 dataset. (a) Preprocessed image. (b) GT. (c) U-Net. (d) U-Net++. (e) Deeplabv3+. (f) Inf-Net. (g) COPLE-Net. (h) CTSA.

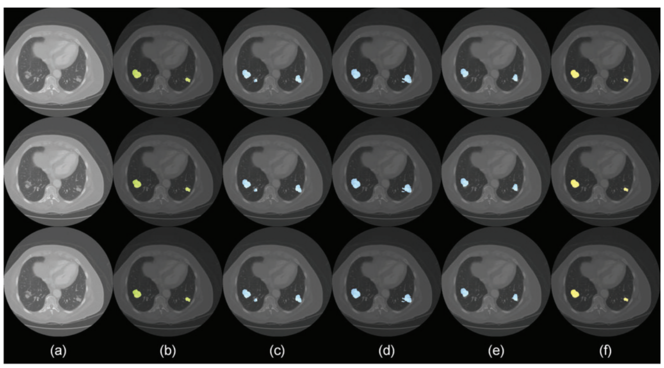

Fig. 7. Visualization results of LIDC-IDRI dataset. (a) Preprocessed image. (b) GT. (c) U-Net. (d) U-Net++. (e) Deeplabv3+. (f) CTSA.

CT sequences and propose a weakly supervised optimization framework based on DRL.

We incorporate OF into the DRL module to enhance the detection and decision-making process. OF effectively captures the motion and structural changes between adjacent slices in a CT sequence, enabling the identification of abnormal pixels. By leveraging this information, the DRL module can focus on areas with significant changes, thereby improving its efficiency in learning an optimal decision path. In addition to this, a key innovation of our method is the design and optimization of the reward function within the DRL module. We emphasize leveraging the continuity between consecutive CT slices to construct the reward function. Specifically, we use the DSC difference between adjacent slices to guide the model in learning a smooth and coherent segmentation strategy, ensuring consistency across slices. This formulation ensures that the model not only optimizes for individual slices but also maintains consistency across the entire sequence, addressing common issues such as fragmented or discontinuous segmentation.

The experimental results demonstrate that the proposed method enhances segmentation accuracy, particularly in detecting subtle lesions. The ablation experiments further validate the contributions of key components, including OF detection, the DRL module, and the constraint segmentation module, each playing a vital role in improving overall performance. Through the runtime analysis, we can see that although our method has a large number of parameters, its running time is also within a reasonable range and is not limited by resource-constrained environments. Compared with the baseline methods, although our method is a weakly supervised framework, it achieves the effect of full supervision through optimization.

Our method uses the most advanced ILD lesion WSS algorithm ViT-MSIL to obtain high-quality initial masks, so as to reduce the impact of inaccurate initial masks on the optimization process. However, the generalization of the model is limited by the sensitivity to the quality of the initial mask, and it cannot be directly applied to datasets from different sources. In future work, we will explore more robust optimization methods, such as introducing GT as an anchor point to constrain the optimization, so as to reduce the impact of the initial masks.

VI. Conclusion

In this work, we propose a novel WSS optimization model, CTSA, a DRL-based lesion segmentation method designed specifically for CT sequences. Our model achieves context-adaptive and environment-interactive segmentation by combining the strengths of a DRL module with a segmentation network. The DRL module is tasked with ensuring the continuity of lesions across consecutive CT slices, by learning the spatial relationships and temporal coherence between adjacent slices. The policy network within the DRL model is trained to strategically locate the optimal region for ILD lesions, and its results are used as prior knowledge to guide the dual U-Net network. This feedback loop allows the dual U-Net to refine the segmentation, ultimately improving the accuracy and robustness of ILD lesion delineation. Our method not only solves the problem of insufficient utilization of spatial information of medical sequences, but also solves the difficulty of data annotation. In the future, we will further optimize the policy network and reward function, explore more efficient segmentation architectures, and extend this model to broader applications, including medical and natural video segmentation tasks.

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